

Assignment – Big Data Processing

1. List 10 Big Data Applications Used in Daily Life

Big Data is used in many applications that we interact with every day. Some common examples are:

1. **Google Search**
 - Google analyzes billions of search queries to provide the most relevant search results in seconds.
 2. **YouTube**
 - YouTube recommends videos based on your watch history, likes, comments, and subscriptions.
 3. **Netflix**
 - Netflix uses Big Data to recommend movies and TV shows according to users' viewing habits.
 4. **Amazon**
 - Amazon suggests products based on browsing history, previous purchases, and customer preferences.
 5. **Instagram**
 - Instagram personalizes the feed and advertisements based on user interactions and interests.
 6. **WhatsApp**
 - WhatsApp processes billions of messages daily and detects spam and suspicious activities using Big Data.
 7. **Google Maps**
 - Google Maps collects live traffic data from millions of users to suggest the fastest routes.
 8. **Swiggy/Zomato**
 - Food delivery apps predict delivery times, recommend restaurants, and optimize delivery routes.
 9. **Online Banking**
 - Banks use Big Data to detect fraudulent transactions and analyze customer spending behavior.
 10. **Healthcare Apps**
 - Hospitals and healthcare applications analyze patient records to improve diagnosis and treatment.
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2. Explain the 5 Vs of Big Data with One Example Each

The **5 Vs** are the main characteristics of Big Data.

1. Volume

Definition:

Volume refers to the huge amount of data generated every second.

Example:

Facebook users upload millions of photos and videos every day.

2. Velocity

Definition:

Velocity means the speed at which data is generated, collected, and processed.

Example:

Stock market systems process thousands of transactions every second.

3. Variety

Definition:

Variety refers to different types of data such as text, images, videos, audio, and sensor data.

Example:

Instagram stores photos, videos, comments, reels, and messages.

4. Veracity

Definition:

Veracity refers to the quality, accuracy, and reliability of data.

Example:

Fake reviews on an e-commerce website must be identified before analyzing customer feedback.

5. Value

Definition:

Value means converting raw data into useful information for better decision-making.

Example:

Amazon analyzes customer purchase history to recommend products, increasing sales.

3. Compare Traditional Data and Big Data

Traditional Data	Big Data
Stores small amounts of data.	Stores extremely large amounts of data (TBs, PBs, and beyond).
Mostly structured data.	Structured, semi-structured, and unstructured data.
Uses relational databases like MySQL and Oracle.	Uses Hadoop, Spark, NoSQL, MongoDB, Cassandra, etc.
Processed on a single server.	Processed on distributed clusters of computers.
Data processing is slower for very large datasets.	Designed for fast and parallel processing.
Limited scalability.	Highly scalable by adding more machines.
Less expensive to manage for small data.	Requires distributed infrastructure but handles massive datasets efficiently.
Suitable for business records and accounting systems.	Suitable for social media, IoT, healthcare, finance, e-commerce, and AI.
Data size is usually in MBs or GBs.	Data size is usually in TBs, PBs, or even EBs.
Examples: Student database, payroll system.	Examples: Google Search, Netflix, Facebook, Amazon.

4. Research One Company and Describe How It Uses Big Data

Company: Netflix

Introduction

Netflix is one of the world's leading online streaming platforms, serving millions of users worldwide. It relies heavily on Big Data to improve customer experience and provide personalized entertainment.

How Netflix Uses Big Data

1. Personalized Recommendations

Netflix analyzes users' watching history, ratings, search history, and viewing time to recommend movies and TV shows.

Example:

If a user watches action movies frequently, Netflix recommends more action-related content.

2. Content Creation

Netflix studies viewers' preferences before producing new shows and movies.

Example:

Viewer data helped Netflix understand the popularity of political dramas, contributing to the production of successful original content.

3. Video Streaming Optimization

Netflix analyzes internet speed and network conditions to automatically adjust video quality, ensuring smooth playback with minimal buffering.

4. Customer Retention

Netflix identifies users who may stop using the platform and recommends engaging content to encourage continued subscriptions.

5. Marketing Strategy

Netflix analyzes customer behavior to send personalized emails, app notifications, and promotional campaigns based on user interests.

Benefits of Big Data for Netflix

- Provides personalized movie and TV show recommendations.
 - Improves customer satisfaction and user experience.
 - Helps create successful original content.
 - Reduces customer churn by increasing engagement.
 - Optimizes streaming quality based on network conditions.
 - Supports better business decisions using data analytics.
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Conclusion

Big Data has become an essential part of modern technology. It helps organizations process massive amounts of information quickly and make better decisions. Applications such as Google, Amazon, Netflix, Swiggy, and healthcare systems use Big Data to improve services, personalize user experiences, detect fraud, optimize operations, and increase efficiency. As data continues to grow, Big Data technologies will play an even more important role in business, education, healthcare, and everyday life.